

1 The Fundamental Laws and Processes of Algebra

Exercises I

- 1. Point out in what sense the usual arrangement of the multiplication of 365 by 492 is an instance of the law of distribution.**

The usual arrangement is

$$\begin{array}{rcl}
 & 365 & \\
 \times & 492 & \\
 \hline
 & 730 & \leftarrow 365 \times 2 \\
 & 3285 & \leftarrow 365 \times 90 \\
 + & 1460 & \leftarrow 365 \times 400 \\
 \hline
 & 179580 &
 \end{array}$$

Decimal notation already gives the multiplier as a sum, one term per digit: $492 = 400 + 90 + 2$. The arrangement therefore computes

$$365 \times 492 = 365 \times (2 + 90 + 400) = 365 \times 2 + 365 \times 90 + 365 \times 400.$$

This is the *law of distribution*, $a(b + c + d) = ab + ac + ad$, with $a = 365$, $b = 2$, $c = 90$, $d = 400$.

The shifting of the second and third rows comes from writing $365 \times 90 = (365 \times 9) \times 10$, an instance of association, together with the decimal place-shift.

- 2. I have a multiplying machine, but the most it can do at one time is to multiply a number of 10 digits by another number of 10 digits. Explain how I can use my machine to multiply 13693456783231 by 46381239245932.**

First we cut the multiplier and multiplicand into digits we can enter into the machine. For decimal notation, we can do this preparation by splitting them into a sum of two

units.

$$13693456783231 = 1369345 \times 10^7 + 6783231$$

$$46381239245932 = 4638123 \times 10^7 + 9245932$$

Now each number is of the form $x_2 \times 10^7 + x_1$, with the two variables consisting of 7 digits each.

Now, giving the multiplicand and multiplier the assignments $a = 1369345$, $b = 6783231$, $c = 4638123$ and $d = 9245932$ and by application of the *law of distribution*

$$\begin{aligned} &13693456783231 \times 46381239245932 \\ &= (a \times 10^7 + b) \times (c \times 10^7 + d) \\ &= a \times 10^7 \times c \times 10^7 + ad \times 10^7 + bc \times 10^7 + bd \\ &= ac \times 10^{14} + (ad + bc) \times 10^7 + bd \end{aligned}$$

Now we can use our multiplying machine to perform the four multiplications, namely ac , ad , bc and bd . The remaining operations, multiplications by a power of 10 and simple addition, can be done by hand.

3. To divide 5004 by 12 is the same as to divide 5004 by 3, and then divide the quotient thus obtained by 4. Of what law of algebra is this an instance?

This is an instance of the *law of association*. Since $12 = 3 \times 4$, we have the statement $5004 \div (3 \times 4) = 5004 \div 3 \div 4$. Notice the removal of the bracket, and since $\div$ precedes the bracket, every sign within has been reversed.

4. If the remainder on dividing N by a be R , and the quotient P , and if we divide P by b and find a remainder S , show that the remainder on dividing N by ab will be $aS + R$. Illustrate with $5015 \div 12$.

Let Q be the quotient on dividing P by b . Then the two divisions are the identities

$$N = aP + R, \quad P = bQ + S, \quad (R < a \text{ and } S < b)$$

Substituting the second in the first, and applying the laws of distribution and association.

$$N = a(bQ + S) + R = abQ + aS + R = ab \times Q + (aS + R)$$

Thus N is a multiple of ab together with $aS + R$. Note that since $R \leq a - 1$ and $S \leq b - 1$,

$$aS + R \leq a(b - 1) + (a - 1) = ab - 1 < ab.$$

Therefore on dividing N by ab the remainder is $aS + R$.

For illustration, take $N = 5015$, $a = 3$ and $b = 4$. Dividing 5015 by 3 gives $P = 1671$, $R = 2$. Dividing 1671 by 4 gives $Q = 417$, $S = 3$. The remainder on dividing 5015 by 12 is $aS + R = 3 \times 3 + 2 = 11$. And we see $5015 = 12 \times 417 + 11$.

5. Show how to multiply two numbers of 10 digits each so as to obtain merely the number of digits in the product, and the first three digits on the left of the product. Illustrate by finding the number of digits, and the first three left-hand digits in the following:

a) $3659893456789325678 \times 342973489379265$;

b) 2^{64} .

Cut each number after its first k digits, so that

$$x = a \times 10^m + r, \quad y = c \times 10^n + s,$$

where a and c have k digits each and the tails satisfy $r < 10^m$, $s < 10^n$. By the law of distribution,

$$xy = ac \times 10^{m+n} + (as \times 10^m + cr \times 10^n + rs).$$

The bracketed part is the whole error made by neglecting the tails, and it is small: since $s < 10^n$, $r < 10^m$, and $rs < 10^{m+n}$,

$$as \times 10^m + cr \times 10^n + rs < (a + c + 1) \times 10^{m+n}.$$

Hence xy lies between $ac \times 10^{m+n}$ and $(ac + a + c + 1) \times 10^{m+n}$. Now ac has $2k - 1$ or $2k$ digits, whereas $a + c + 1$ has at most $k + 1$; so the correction $a + c + 1$ can disturb only the last $k + 1$ digits of ac , and the leading $k - 2$ or so digits, together with the number of digits, are left untouched. Thus the number of digits in xy is the number of digits in ac plus $m + n$, and the first digits of xy are the first digits of ac as long as the addition of $a + c + 1$ does not carry into the desired digits, if it does we can take k to be larger.

Illustration (a). Take $k = 5$:

$$3659893456789325678 = 36598 \times 10^{14} + 93456789325678,$$

$$342973489379265 = 34297 \times 10^{10} + 3489379265.$$

Then $ac = 36598 \times 34297 = 1255201606$, and $a + c + 1 = 70896$, so

$$1255201606 \times 10^{24} < xy < 1255272502 \times 10^{24}.$$

Both bounds have $10 + 24 = 34$ digits and begin with 1255. The product has therefore 34 digits, and its first three digits are 125.

Illustration (b). Since $2^{64} = (2^{32})^2$ and $2^{32} = 4294967296$ has exactly ten digits, this is a product of two numbers of ten digits each. With $k = 5$,

$$4294967296 = 42949 \times 10^5 + 67296,$$

so $ac = 42949^2 = 1844616601$ and $a + c + 1 = 85899$, giving

$$1844616601 \times 10^{10} < 2^{64} < 1844702500 \times 10^{10}.$$

Both bounds have $10 + 10 = 20$ digits and begin with 1844. Hence 2^{64} has 20 digits, the first three being 184.

6. Express in the simplest form—

$$-(-(-(-(\dots (-1) \dots)))),$$

a) where there are $2n$ brackets;

b) where there are $2n + 1$ brackets; n being any whole number whatever.

Let N be the number of brackets. Each of the N brackets is preceded by a $-$ sign, and by the law of association, removing such a bracket reverses the sign of its content. Since the innermost bracket contains -1 , the value is $(-1)^N \times (-1) = -(-1)^N$.

a) If $N = 2n$, then

$$\begin{aligned} -(-1)^N &= -(-1)^{2n} \\ &= -((-1)^2)^n \\ &= -(1^n) \\ &= -1 \end{aligned}$$

b) If $N = 2n + 1$, then

$$\begin{aligned} -(-1)^N &= -(-1)^{2n+1} \\ &= -[(-1)^{2n} \times (-1)] \\ &= -[(+1) \times (-1)] \\ &= -(-1) \\ &= +1 \end{aligned}$$

7. Simplify and condense as much as possible—

$$2a - \{3a - [a - (b - a)]\}.$$

$$\begin{aligned}
2a - \{3a - [a - (b - a)]\} &= 2a - \{3a - [a - b + a]\} \\
&= 2a - \{3a - [2a - b]\} \\
&= 2a - \{3a - 2a + b\} \\
&= 2a - \{a + b\} \\
&= 2a - a - b \\
&= a - b
\end{aligned}$$

8. Simplify—

a) $3\{4 - 5[6 - 7(8 - 9 \cdot \overline{10 - 11})]\};$

b) $\frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7}(\frac{1}{8} - \frac{1}{9} \cdot \overline{\frac{1}{10} - \frac{1}{11}})]\}.$

$$\begin{aligned}
a) \ 3\{4 - 5[6 - 7(8 - 9 \cdot \overline{10 - 11})]\} &= 3\{4 - 5[6 - 7(8 - 9 \cdot -1)]\} \\
&= 3\{4 - 5[6 - 7(8 + 9)]\} \\
&= 3\{4 - 5[6 - 7(17)]\} \\
&= 3\{4 - 5[6 - 119]\} \\
&= 3\{4 - 5[-113]\} \\
&= 3\{4 + 565\} \\
&= 3\{569\} \\
&= 1707
\end{aligned}$$

$$\begin{aligned}
b) \ \frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7}(\frac{1}{8} - \frac{1}{9} \cdot \overline{\frac{1}{10} - \frac{1}{11}})]\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7}(\frac{1}{8} - \frac{1}{9} \cdot \frac{1}{110})]\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7}(\frac{1}{8} - \frac{1}{990})]\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7} \cdot \frac{491}{3960}]\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{491}{27720}]\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{1}{5} \cdot \frac{4129}{27720}\} \\
&= \frac{1}{3}\{\frac{1}{4} - \frac{4129}{138600}\} \\
&= \frac{1}{3} \cdot \frac{30521}{138600} \\
&= \frac{30521}{415800}
\end{aligned}$$

9. Simplify—

$$1 - (2 - (3 - (4 - \dots (9 - (10 - 11)) \dots))).$$

Since every bracket is preceded by a $-$ sign, by the law of association each bracket may be removed if every sign within it is reversed, giving

$$1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 - 10 + 11$$

Grouping in pairs,

$$(1 - 2) + (3 - 4) + (5 - 6) + (7 - 8) + (9 - 10) + 11 = 5 \times (-1) + 11 = 6.$$

10. Distribute the following products:

a) $(a + b) \times (a + b);$

b) $(a - b) \times (a + b);$

c) $(3a - 6b) \times (3a + 6b);$

d) $(\frac{1}{3}a - \frac{1}{6}b) \times (\frac{1}{3}a + \frac{1}{6}b).$

a) $(a + b) \times (a + b) = (a + b)a + (a + b)b = aa + ba + ab + bb = a^2 + 2ab + b^2.$

b) $(a - b) \times (a + b) = (a - b)a + (a - b)b = aa - ba + ab - bb = a^2 - b^2.$

c) $(3a - 6b) \times (3a + 6b) = 9aa + 18ab - 18ab - 36bb = 9a^2 - 36b^2.$

d) $(\frac{1}{3}a - \frac{1}{6}b) \times (\frac{1}{3}a + \frac{1}{6}b) = \frac{1}{9}aa + \frac{1}{18}ab - \frac{1}{18}ab - \frac{1}{36}bb = \frac{1}{9}a^2 - \frac{1}{36}b^2.$

11. Simplify, by expanding and condensing as much as possible—

$$\begin{aligned} & \{(m + 1)a + (n + 1)b\}\{(m - 1)a + (n - 1)b\} \\ & + \{(m + 1)a - (n + 1)b\}\{(m - 1)a - (n - 1)b\}. \end{aligned}$$

Let $A = (m + 1)a$, $B = (n + 1)b$, $C = (m - 1)a$, $D = (n - 1)b$. Then

$$\begin{aligned} & \{(m + 1)a + (n + 1)b\}\{(m - 1)a + (n - 1)b\} \\ & + \{(m + 1)a - (n + 1)b\}\{(m - 1)a - (n - 1)b\} \\ & = (A + B)(C + D) + (A - B)(C - D) \\ & = (A + B)C + (A + B)D + (A - B)C - (A - B)D \\ & = AC + BC + AD + BD + AC - BC - AD + BD \\ & = 2AC + 2BD \\ & = 2(m + 1)a(m - 1)a + 2(n + 1)b(n - 1)b \\ & = 2a^2(m + 1)(m - 1) + 2b^2(n + 1)(n - 1) \\ & = 2a^2(m^2 - 1) + 2b^2(n^2 - 1), \end{aligned}$$

collecting the factors by commutation in the second-last line, and using the result of Exercise 10 (b) in the last.

12. Simplify—

$$\left(x + \frac{1}{x}\right)\left(y + \frac{1}{y}\right) + \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right).$$

This exercise has the same form as exercise 11, but with $A = x$, $B = \frac{1}{x}$, $C = y$, $D = \frac{1}{y}$. So we can start from the expansion obtained there,

$$(A + B)(C + D) + (A - B)(C - D) = 2AC + 2BD = 2xy + 2\frac{1}{xy}$$

13. Simplify—

$$\left(x + \frac{1}{x}\right)\left(y + \frac{1}{y}\right)\left(z + \frac{1}{z}\right) - \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)\left(z - \frac{1}{z}\right).$$

Let $A = x$, $B = \frac{1}{x}$, $C = y$, $D = \frac{1}{y}$, $E = z$, $F = \frac{1}{z}$, then the expression is

$$(A + B)(C + D)(E + F) - (A - B)(C - D)(E - F).$$

Multiplying the two-factor expansions of exercise 11 by the third factor,

$$\begin{aligned} (A + B)(C + D)(E + F) &= \\ (AC + AD + BC + BD)(E + F) &= \\ ACE + ADE + BCE + BDE + ACF + ADF + BCF + BDF, \\ (A - B)(C - D)(E - F) &= \\ (AC - AD - BC + BD)(E - F) &= \\ ACE - ADE - BCE + BDE - ACF + ADF + BCF - BDF. \end{aligned}$$

Then we have,

$$\begin{aligned} (A + B)(C + D)(E + F) - (A - B)(C - D)(E - F) &= \\ ACE + ADE + BCE + BDE + ACF + ADF + BCF + BDF - \\ (ACE - ADE - BCE + BDE - ACF + ADF + BCF - BDF) &= \\ 2ADE + 2BCE + 2ACF + 2BDF &= \\ 2\frac{xz}{y} + 2\frac{yz}{x} + 2\frac{xy}{z} + 2\frac{1}{xyz} \end{aligned}$$

14. Expand and condense as much as possible—

$$(\frac{1}{2}x + \frac{1}{4}y + \frac{1}{8}z)(\frac{1}{2}x - \frac{1}{4}y + \frac{1}{8}z).$$

Let $P = \frac{1}{2}x + \frac{1}{8}z$, then, the expression is

$$(P + \frac{1}{4}y)(P - \frac{1}{4}y) = PP - \frac{1}{16}yy$$

by the result of Exercise 10 (b). By that of 10 (a),

$$PP = (\frac{1}{2}x + \frac{1}{8}z)(\frac{1}{2}x + \frac{1}{8}z) = \frac{1}{4}xx + 2 \cdot \frac{1}{2} \cdot \frac{1}{8}xz + \frac{1}{64}zz,$$

so the expression equals

$$\frac{1}{4}x^2 + \frac{1}{8}xz - \frac{1}{16}y^2 + \frac{1}{64}z^2.$$